

Optimal Sizing of a Nuclear Reactor for Embedded Grid Systems

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ILLINOIS



Outline

1 Motivation

Illinois Climate Action Plan (iCAP)

Need for Nuclear

Framing the Question

2 Methods

Overview

Methods for RAVEN

Methods for Temoa

3 Grid Characterization: RAVEN

4 Optimal Sizing: Temoa

Temoa: Business As Usual

Temoa: Nuclear Scenarios

Scenario 1: Zero Capital Costs

Scenario 2: No Capacity Limit

Scenario 3: Small Modular Reactor

5 Conclusion

6 Future Work

iCAP Goal and Obstacles

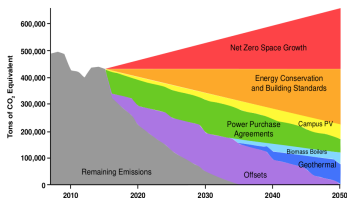


Figure: Shows projected CO₂ emissions for UIUC [9]. Offsets include shutdown of the Blue Waters Supercomputer.

Goal:

Carbon neutrality by 2050 or sooner.

Obstacles:

- 1 Requires *zero net space growth*.
- 2 Campus depends on a system of steam tunnels for heating.
- 3 and more...



The Nuclear Option

Nuclear energy...

- ① ...produces almost no carbon emissions [8].
- ② ...can produce high-temperature steam.
- ③ ...requires little physical space*.

*compared to solar and wind.

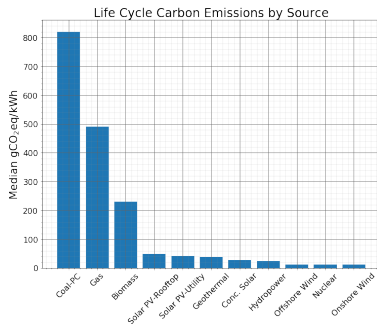


Figure: Lifetime carbon-equivalent emissions by energy source from IPCC findings [8].

Small Modular and Micro-reactors



- 20 MWth (micro) to 300 MWth (small modular)
- Fewer resource requirements (area, shielding, operations)



Features:

- Factory fabricated
- Transportable (especially for micro-reactors)
- Walk-away safe
- Potential for dispatchability

Figure: Transportable reactor concept. Image reproduced from US-DOE Nuclear Fast Facts [11].

What is the optimal size for a nuclear reactor on the UIUC grid?

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To answer this question we considered two modeling approaches:

- ① RAVEN (INL) - Risk Analysis and Virtual Environment [2][6]
- ② TEMOA (NCSU) - Tools for Energy Model Optimization and Analysis [3][4][7]

Both modeling tools are open source and use publicly available version control software, `Git`, to track changes.

The analysis in RAVEN requires some external modules that are not currently available to the public.

Workflow in RAVEN

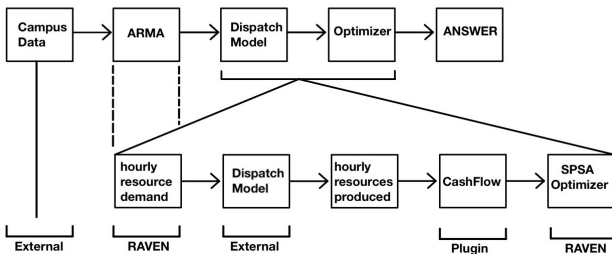


Figure: A general optimization workflow in RAVEN. Only the ARMA step was used to characterize the UIUC grid.

Temoa Implementation



Temoa uses linear optimization to search decision space [7].

- ① Objective Function (minimizes system cost)
- ② Constraints
 - ① Demand must be satisfied at each time step (always).
 - ② Carbon limits must be satisfied at each time step (optionally).
- ③ Variables
 - ① Cost
 - ② Generation
 - ③ Capacity

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Step 1: Generate Synthetic Histories

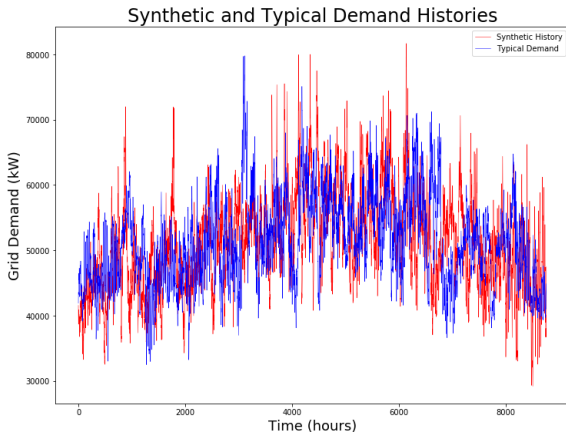


Figure: Shows the synthetic (red) vs typical (blue) hourly electricity demand at UIUC.

Step 1: Generate Synthetic Histories (continued)

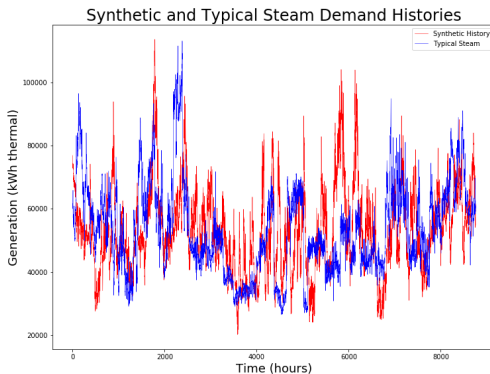


Figure: Shows the synthetic (red) vs typical (blue) hourly steam demand at UIUC.

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BAU: Grid Model

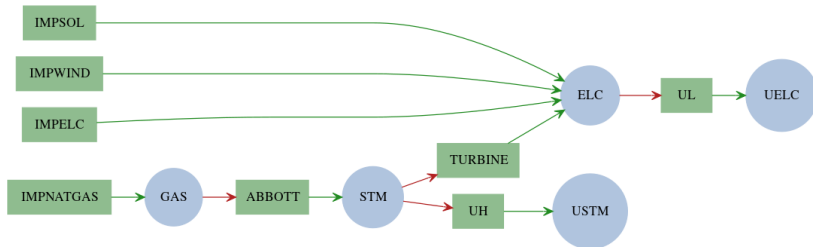


Figure: Graph representation of the UIUC embedded grid.

BAU: Generation

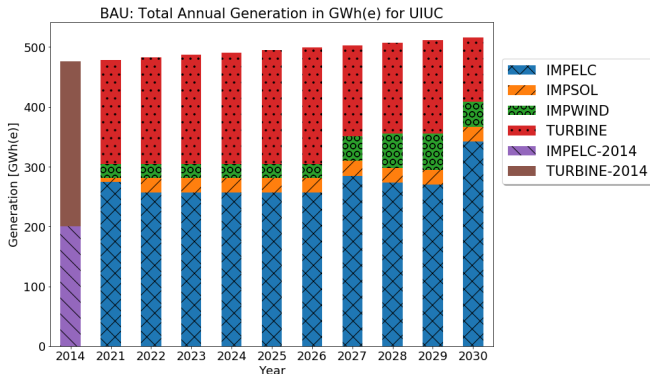


Figure: The change in activity from each energy source from 2020-2030. Assuming 1% demand growth each year

BAU: Emissions

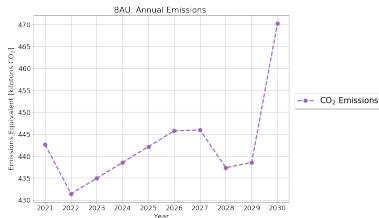


Figure: The change in activity from each energy source from 2020-2030. Assuming 1% demand growth each year

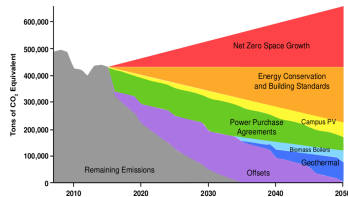


Figure: Predicted growth in emissions from iCAP [9].

Nuclear Scenarios



- ① Scenario 1: Zero Capital Costs
- ② Scenario 2: No Capacity Limit
- ③ Scenario 3: Limited to Small Modular Reactor (100MWth)

Table: Summary of Nuclear Scenarios. Costs from EIA and NEI reports [5][10]. Assumes thermal efficiency of 33%.

Scenario	Operation Costs [\$ /MWth(th)]	Capital Costs [M\$/MWth]	Maximum Capacity [MWth]
1	8.91	-	-
2	8.91	1.982	-
3	8.91	1.982	100

Assumptions



- ① Assumes fixed capital and variable costs throughout the model time horizon.
- ② Nuclear Reactor
 - Assumes LWR due to availability of cost estimates.
 - Assumes 92% capacity factor.
- ③ Wind Power
 - Assumes 31% capacity factor.
- ④ Solar Power
 - Assumes capacity factor of 16.8%, based on the UIUC solar farm data [1].

Nuclear Scenarios: Grid Model

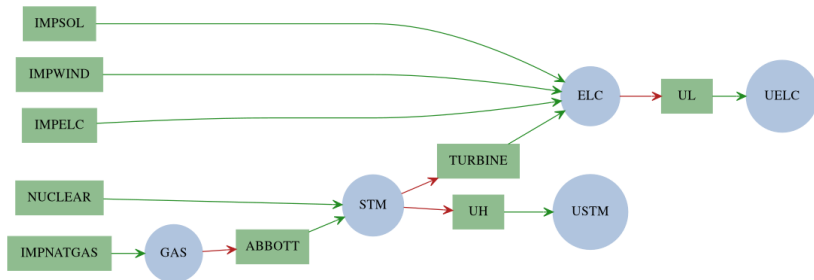


Figure: Graph representation of the UIUC grid with nuclear reactor.

Scenario 1: Generation

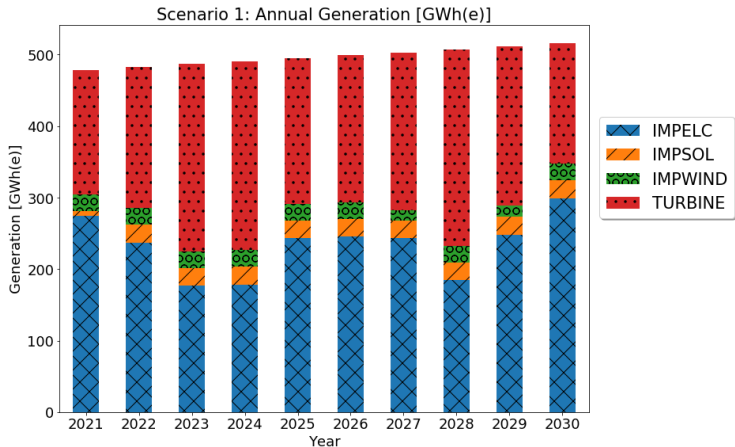


Figure: The electric generation without a cost constraint on nuclear

Scenario 1: Emissions



Figure: The carbon equivalent emissions without a cost constraint on nuclear

Scenario 2: Generation

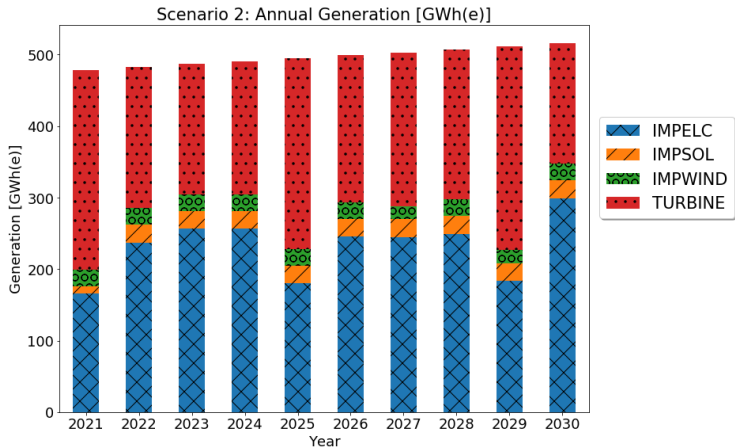


Figure: The electric generation without a size constraint on nuclear

Scenario 2: Emissions

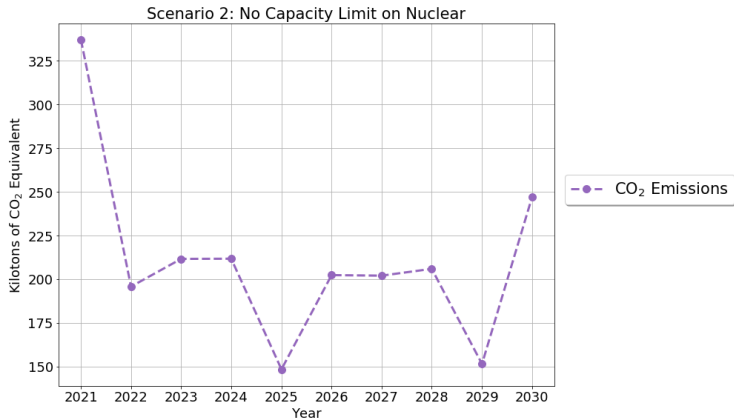


Figure: The carbon equivalent emissions without a size constraint on nuclear

Scenario 3: Generation

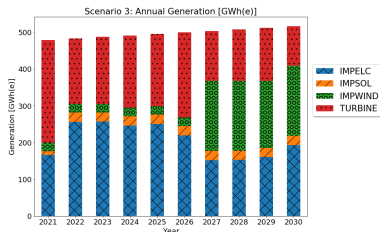


Figure: The electric generation with constrained nuclear.

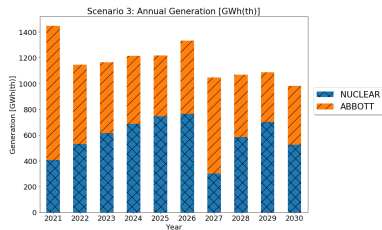


Figure: The steam generation with constrained nuclear

Scenario 3: Generation

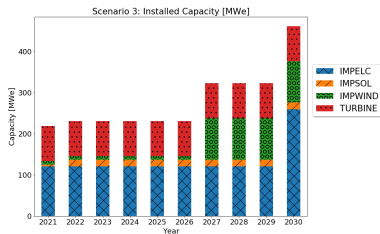


Figure: The electric capacity with constrained nuclear.

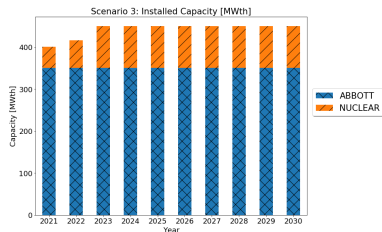


Figure: The steam capacity with constrained nuclear.

Scenario 3: Emissions

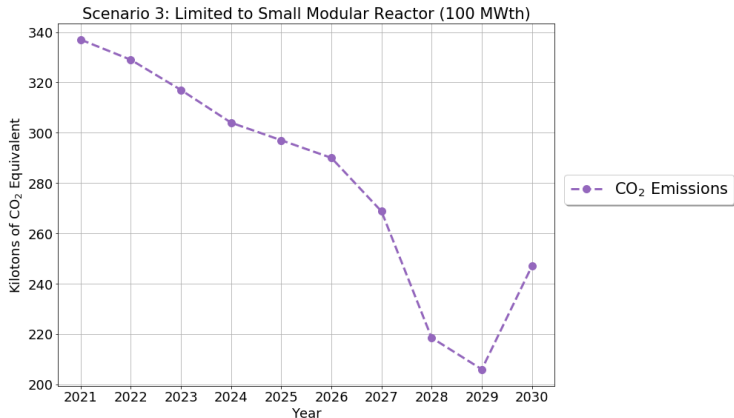


Figure: The carbon equivalent emissions without a cost constraint on nuclear

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Conclusion



- ① Replacing “ABBOTT” with nuclear would resolve all of the Universities carbon goals, *regardless of other offsets and building growth.*
- ② Adding, even limited, nuclear capacity will cost effectively meet carbon goals until mid-decade.
- ③ This model is agnostic to implementation:
 - One 100 MWth small modular reactor
 - Series of 20 MWth micro-reactors

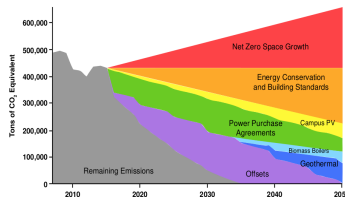


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Future Work



- ① Explore parametric uncertainty in Temoa with
 - Monte Carlo sampling
 - Stochastic Optimizationparticularly for natural gas prices and nuclear capital costs.
- ② Explore structural uncertainty in Temoa using Modeling-to-Generate-Alternatives.
- ③ Explore unmodeled markets and potential cash flows by developing a dispatch model for RAVEN.

Acknowledgement

This work was made possible with data provided by UIUC Facilities and Services, in particular, Morgan White, Mike Marquissee, and Mike Larson.

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<https://www.energy.gov/ne/downloads/ultimate-fast-facts-guide-nuclear-energy>.

Questions?

Mathematics of Temoa

Minimize

$$C_{total} = \sum_{t,v} C_{t,v} \quad (1)$$

Subject to

$$D_{s,p} = \sum_{s,p} G_{s,p}$$

$$L_p = \sum_{t,p} \hat{R}_{CO_2,\{t,p\}}$$

Where

C_{total} = total cost

$D_{s,p}$ = energy demand by sector, time period

$G_{s,p}$ = energy generation by sector, time period

L_p = emission limits by time period

$\hat{R}_{CO_2,\{t,p\}}$ = emissions by technology, time period

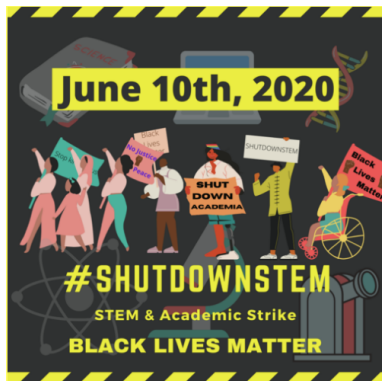
- *What does "TURBINE" mean?*

The "TURBINE" technology simply converts steam to electricity since Abbott power plant is actually a cogeneration plant. We assumed that a nuclear reactor that could replace Abbott would also be used for cogeneration.

- *Can this analysis be applied to other universities or energy systems?*

Yes. While the University of Illinois is unique in its self-reliance, the idea that nuclear power fulfills a role in the energy mix that is not easily satisfied by renewables is not.

#ShutDownSTEM



This lecture has been pre-recorded.
Questions can be directed to:
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Thank you.